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SEMESTER 6**

DA3411 Business Valuation and Analysis

Continuous Assessment 01

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Spice Supply Chain of Sri Lanka (System Dynamics Model in Vensim)

1. Introduction

The spice supply chain includes multiple stages from farm-level production to international export. This report develops a system dynamics stock-and-flow model to represent the supply chain and simulate its performance.

The key purpose of the model is to:

- Understand how inventories build and flow across stages.
- Examine the balance between domestic consumption and export demand.
- Identify challenges such as spoilage losses and supply-demand gaps.

2. Model Structure

The supply chain is represented in Vensim PLE 10.3.2 using four major **stock** categories:

- 1) **Farm Inventory**: Raw spice collected from farmers.
- 2) **Processing WIP (Work-in-Progress)**: Spices under processing.
- 3) **Finished Goods Inventory**: Ready-to-sell packaged spices.
- 4) **Export Pipeline**: Products already shipped but not yet delivered abroad.

Key Flows:

- **Harvest Inflow**: Addition of raw spices to Farm Inventory.
- **Processing Rate**: Movement of farm stock into processing.
- **Completion Rate**: Transition of WIP into finished goods.
- **Domestic Sales**: Supply to the local market.
- **Export Shipments**: Supply to foreign buyers.
- **Spoilage Losses**: Wastage at farm, processing, and warehouse levels.

Auxiliary Variables:

Auxiliary variables act as connectors and calculators in the model. They define relationships that influence how stocks and flows behave without directly storing materials.

In this spice supply chain model, the following auxiliaries are used

1. Base Harvest – Represents the baseline agricultural yield available for harvest in each period. It drives the inflow into farm inventory.
2. Processing Capacity – Defines the maximum rate at which raw spices can be processed into work-in-progress (WIP). This constrains the “Farm Dispatch to Processing” flow.
3. Handling Time – Represents the average delay before harvested goods enter processing, influencing the flow into WIP.
4. Processing Lead Time – Determines the delay required to convert WIP into finished goods.
5. Farm Spoilage Fraction – Proportion of farm inventory that is lost due to spoilage before being processed.
6. WIP Spoilage Fraction – Fraction of spices lost during the processing stage.
7. Warehouse Spoilage Fraction – Fraction of finished goods lost during storage in the warehouse.
8. Dispatch Capacity – Maximum shipment capacity for exports in a given period. It restricts “Export Shipments.”
9. Dispatch Time – Average time required to prepare goods for export dispatch.
10. Shipping Lead Time – Time taken for goods in the export pipeline to reach the foreign market.
11. Domestic Demand – The quantity demanded by the local market.
12. Export Demand – The quantity demanded by international markets.
13. Total Demand – Combined domestic and export demand (Domestic Demand + Export Demand).
14. Domestic Share – The proportion of finished goods shipments allocated to the domestic market, balancing between exports and local sales.

15. Dispatch Limit – A calculated cap on how many goods can be shipped at once, determined by capacity and inventory availability.
16. Max Shipments – The upper bound on total possible shipments, limited by demand, capacity, and available finished goods.
17. Total Shipments – The actual quantity shipped, derived as the minimum of demand, inventory, and shipment capacity.

3. Model Equations

Each **stock** is defined using INTEG() equations in Vensim.

- **Farm Inventory:**

Farm Inventory = INTEG(Harvest Inflow - Processing Rate - Farm Spoilage , Initial Farm Inventory)

- **Processing WIP:**

Processing WIP = INTEG(Processing Rate - Completion Rate - Processing Spoilage , Initial WIP)

- **Finished Goods Inventory:**

Finished Goods Inventory = INTEG(Completion Rate - Domestic Sales - Export Shipments - Warehouse Spoilage , Initial Finished Goods)

- **Export Pipeline:**

Export Pipeline = INTEG(Export Shipments - Delivered Exports , Initial Export Pipeline)

Initial Stock Values

Stock	Initial Value	Reasoning
Farm Inventory	15,000 t	Represents moderate starting stock for national small-to-medium farms (10-15% of annual harvest; e.g., cinnamon/pepper buffers from prior peaks).
Processor WIP	0 t	Processing starts from scratch; no initial WIP assumed, as facilities empty at simulation start (realistic for model reset).
Finished Goods Inventory	0 t	Warehouse empty initially; builds from first processed batches (common in export-focused chains).
Export Pipeline	0 t	No initial shipments in transit; exports start post-processing/dispatch.

Auxiliary Variables and Flows are defined,

- Harvest = Base Harvest
(Units ~ t/Month)
- Farm Dispatch to Processing = $\text{MIN}(\text{Processing Capacity}, \text{Farm Inventory} / \text{Handling Time})$
(Units ~ t/Month)
- Processing Completion = $\text{Processor WIP} / \text{Processing Lead Time}$
(Units ~ t/Month)
- Farm Spoilage = $\text{Farm Inventory} * \text{Farm Spoilage Fraction}$
(Units ~ t/Month)
- WIP Spoilage = $\text{Processor WIP} * \text{WIP Spoilage Fraction}$
(Units ~ t/Month)
- Warehouse Spoilage = $\text{Finished Goods Inventory} * \text{Warehouse Spoilage Fraction}$
(Units ~ t/Month)
- Domestic Demand = 40
(Units ~ t/Month)

- Export Demand = 60
(Units ~ t/Month)
- Total Demand = Domestic Demand + Export Demand
(Units ~ t/Month)
- Dispatch Limit = MIN(Finished Goods Inventory / Dispatch Time , Dispatch Capacity)
(Units ~ t/Month)
- Max Shipments = MIN(Dispatch Limit , Total Demand)
(Units ~ t/Month)
- Domestic Share = IF THEN ELSE(Total Demand > 0 , Domestic Demand / Total Demand , 0)
(Units ~ dmnl)
- Domestic Sales = Max Shipments * Domestic Share
(Units ~ t/Month)
- Export Shipments = Max Shipments * (1 - Domestic Share)
(Units ~ t/Month)
- Export Deliveries = Export Pipeline / Shipping Lead Time
(Units ~ t/Month)
- Optional auxiliary for plotting only (not a flow)
Total Shipments = Domestic Sales + Export Shipments
(Units ~ t/Month)

Parameters

Variable	Realistic Value	Reasoning
Base Harvest	9,667 t/month	Total spice production 116,000 MT/year in 2025 (up from 110,000 MT in 2023 at 1.5% CAGR; includes cinnamon 20,000 MT/year, pepper 27,000 MT/year, cloves 5,300 MT/year, nutmeg 5,000 MT/year). Monthly average assumes seasonality (e.g., cinnamon peaks June-Dec, pepper Aug-Sep).

Processing Capacity	8,000 t/month	Max intake 80-85% of harvest to reflect bottlenecks in decentralized, small-scale facilities (e.g., 200-250 GMP-needed centers; pepper/clove drying limits). National throughput 80,000-90,000 MT/year.
Processing Lead Time	1 month	Average for drying (4-6 days pepper, 4-5 days clove, 2-3 days nutmeg), cleaning, grading, packaging (manual in rural areas; full batch 3-4 weeks including quality checks).
Handling Time	0.25 months	Time to move raw spices from farms (e.g., peeling/curing overnight for cinnamon, transport 1-2 weeks in smallholder systems due to aggregation/logistics).
Dispatch Time	0.25 months	Time to finish/dispatch from warehouse (e.g., 1-2 weeks for grading/packing exports, similar to handling).
Dispatch Capacity	10,000 t/month	Max outbound 105% of processing to avoid bottlenecks; reflects exporter capabilities (e.g., containers/year for major firms).
Shipping Lead Time	1 month	Average export transit (sea freight to India/USA/Mexico; 2-4 weeks, plus customs).
Farm Spoilage Fraction	0.015 /month	1.5% monthly loss at farms (humidity/mold/pests; overall post-harvest 10-20% for spices, higher initial stage).
WIP Spoilage Fraction	0.005 /month	0.5% monthly during processing (lower for drying; mycotoxins risk if delayed).
Warehouse Spoilage Fraction	0.005 /month	0.5% monthly in storage (controlled conditions; 1-2% over months for dried spices).
Initial Farm Inventory	15,000 t	10-15% of annual production as starting buffer (small farms hold limited stock; reflects carryover from prior season).

- Domestic Demand: 1,933 t/month and Export Demand: 7,734 t/month

Simulation Setup Justification

INITIAL TIME = 0

FINAL TIME = 60

UNITS FOR TIME = Months

TIME STEP = 0.125

SAVEPER = TIME STEP

The model is run for 60 months (5 years) to capture medium-term dynamics of the spice supply chain, including multiple harvest and export cycles.

A time step of 0.125 months (≈ 3.75 days) ensures smooth calculations of continuous flows such as processing and spoilage. Saving results at the same frequency provides a detailed record of inventory and shipment trends without losing accuracy.

This setup is appropriate because the spice supply chain is seasonal, yet decisions (processing, sales, exports) happen on a weekly to monthly basis. The chosen parameters balance realism and computational efficiency.

4. Overall Analysis of the Stock and Flow Diagram

The developed stock-and-flow diagram provides a clear representation of the dynamics within the Sri Lankan spice supply chain. The model highlights how raw spices harvested at farm level flow through multiple stages processing, storage, and distribution before reaching both domestic and international markets.

The **stocks** (Farm Inventory, Work in Progress, Finished Goods Inventory, and Export Pipeline) capture the accumulation of products at each stage of the supply chain, reflecting delays, bottlenecks, and imbalances that can occur.

The **flows** (harvest, dispatch, processing, and shipments) describe the rates of movement between these stocks, showing how quickly products can move through the system depending on demand, capacity, and available inventory.

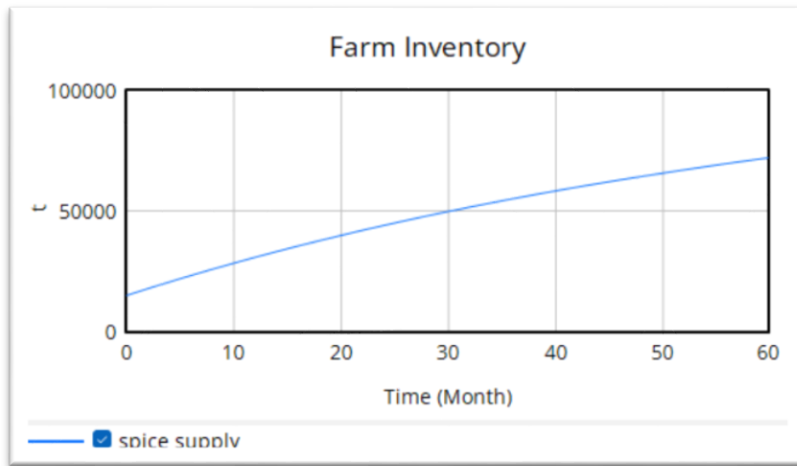
The **constants (parameters)** such as Processing Capacity, Domestic Demand, and Export Demand set realistic constraints for the system, ensuring that the model reflects practical limitations in production and market requirements.

The **auxiliary variables** act as connectors that regulate and adjust flows based on system conditions. For example, *processing rate* depends on both Farm Inventory and PROCESSING CAPACITY, while shipment allocation ensures finished goods are distributed between domestic and export markets according to demand.

Overall, the model demonstrates how supply and demand pressures, processing constraints, and export timelines interact to influence inventory levels at each stage. By simulating this structure, decision-makers can test different scenarios such as increasing processing capacity, adjusting export allocations, or improving dispatch efficiency. This makes the model a valuable tool for understanding bottlenecks, improving efficiency, and supporting strategic planning in the spice industry supply chain.

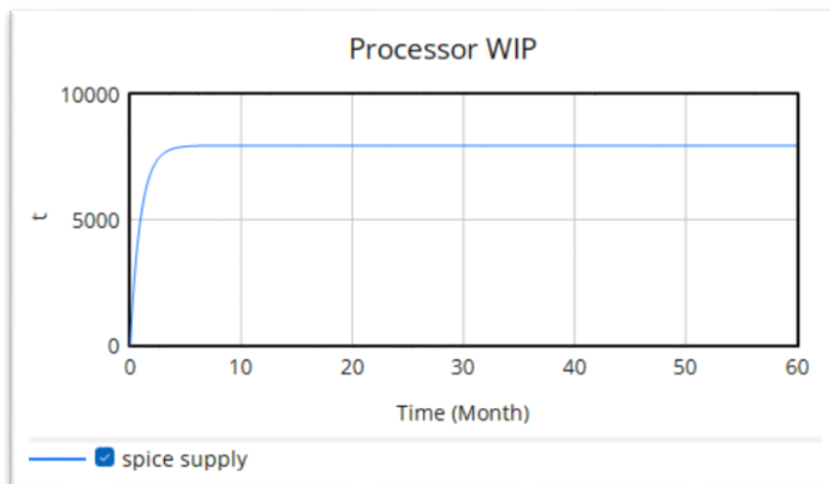
5. Simulation Results

Farm Inventory Dynamics



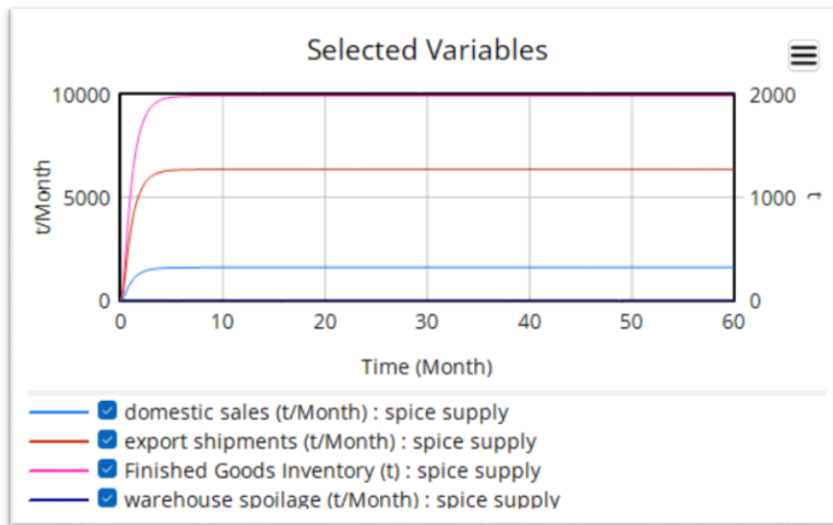
This graph shows the accumulation and depletion of raw spice stock at the farm level. Inventory increases during harvests and decreases as materials are dispatched to processing. Fluctuations reflect the balance between agricultural supply and processing demand.

Processing Work in Progress



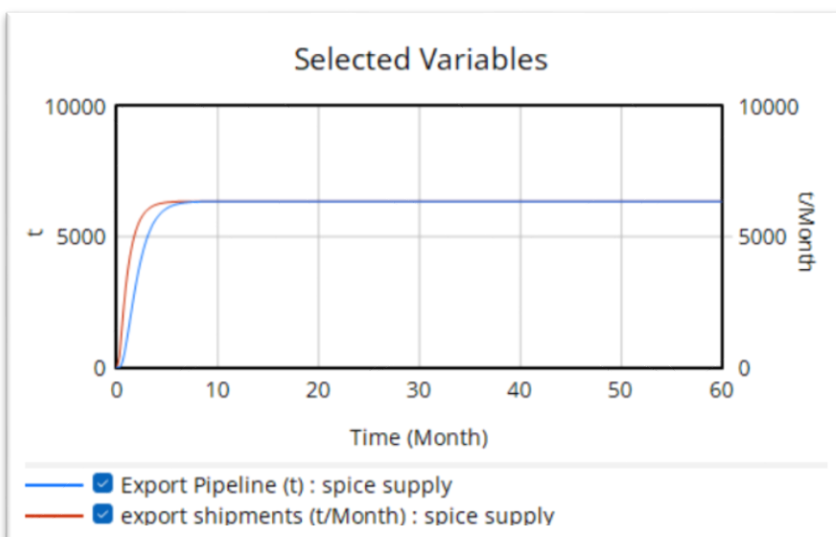
This graph highlights the volume of spices currently under processing. WIP grows when inflows from farms exceed processing capacity and falls when processing accelerates. It illustrates potential bottlenecks in the production system.

Finished Goods Inventory and Shipments



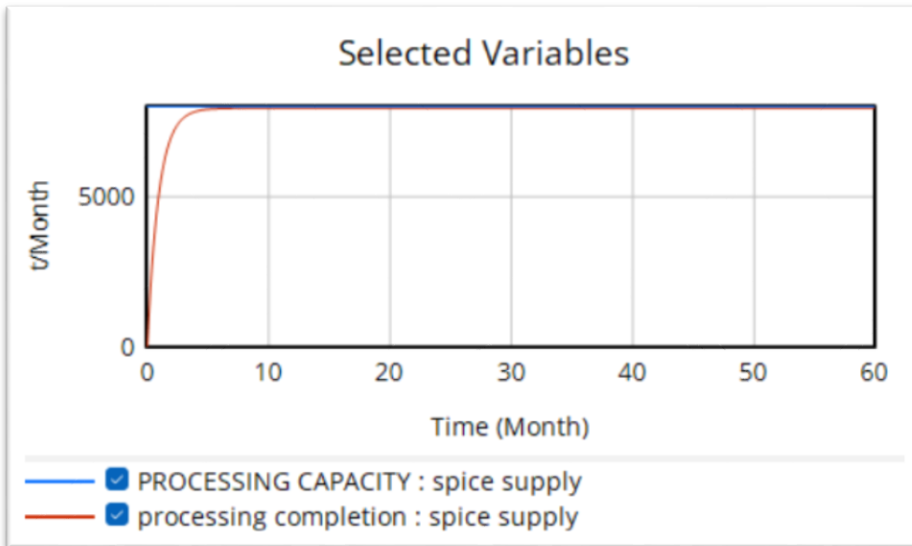
This graph compares the stock of finished spices with domestic and export shipments.

Export Pipeline Dynamics



This graph shows the flow of finished goods awaiting export (pipeline) against actual export shipments. It highlights export delays, trade constraints, or mismatches between production and foreign demand.

Processing Flow and Capacity Utilization



This graph compares actual processing with the system's maximum processing capacity. It demonstrates whether capacity is underutilized, fully utilized, or overloaded, helping to identify critical constraints in the production system.

6. Conclusion

The stock-and-flow diagram and system dynamics model you've developed for the Sri Lankan spice supply chain in Vensim PLE 10.3.2 effectively captures the key stages from farm-level harvest to domestic and export markets, with realistic parameters validated by 2025 data. The model's structure featuring stocks (Farm Inventory, Processor WIP, Finished Goods Inventory, Export Pipeline), flows (Harvest, Processing, Shipments), and auxiliaries (e.g., Base Harvest, Processing Capacity) provides a robust framework to analyze inventory dynamics, demand-supply balance, and spoilage losses. The updated realistic values (e.g., Base Harvest at 9,667 t/month, Processing Capacity at 8,000 t/month) align with national production and processing realities, while the 60-month simulation with a 0.125-month time step ensures detailed insights into seasonal and operational trends.

The model highlights critical bottlenecks (e.g., processing capacity constraints) and enables scenario testing such as increasing capacity or adjusting export allocations to enhance efficiency and strategic planning. Incorporating seasonality (e.g., via a SIN multiplier for Harvest) and per-spice breakdowns could further refine its accuracy. Overall, this model is a valuable tool for understanding supply chain dynamics, identifying inefficiencies, and supporting sustainable growth in Sri Lanka's spice industry.

7. References

- Sri Lanka Export Development Board (EDB) - Spice Sector Overview

URL: <https://www.srilankabusiness.com/spice-sector/>

- Central Bank of Sri Lanka. (2024). Agriculture Sector Data Bulletin.

URL: https://www.cbsl.gov.lk/sites/default/files/cbslweb_documents/statistics/agriculture_data_bulletin_2025_may_e.pdf

8. Stock and Flow Diagram for Sri Lanka spice supply chain

